complicates the error source analysis of deployed PFNs.

Our Contribution In this work, we make the following contributions to study the sources of error and uncertainty in PFNs with a focus on amortization:

1. A decomposition of the error of any PFN under a deployment distribution, splitting into prior, context and amortization error.
2. A study of the amortization error of PFNs under its training prior, including a comparison to a realistic reference PPD (Section 2).
3. An extended investigation of the amortization error of PFNs under a deployment task distribution (Section 3).
4. An analysis of approximate Bayesian model averages over PFN weights (Section 4): Deep ensembles consistently improve predictions, whereas sampling and variational learning add little.

2 AMORTIZATION ERROR

2.1 Setup

Following Nagler (2023), a *task* is a law p over $(\mathbf{X}, Y) \in \mathcal{X} \times \mathcal{Y}$ with covariate law $p_{\mathbf{X}}$ and conditional $p(y \mid \mathbf{x})$, and a dataset $\mathcal{D}_n = \{(\mathbf{X}_i, Y_i)\}_{i=1}^n$ consisting of n independent and identically distributed (iid) draws from it. A PFN assumes a *prior* Π over pairs (p, N) of a task and a context size, whose marginal over N is the *size distribution* Ψ. Given $(p, N{=}n)$, the context is $\mathcal{D}_n \sim p^{\otimes n}$, and the PFN targets the posterior predictive distribution (PPD)

$$\pi_\Pi(y \mid \mathbf{x}, \mathcal{D}_n) = \int p(y \mid \mathbf{x})\, d\Pi(p \mid \mathbf{x}, \mathcal{D}_n). \tag{1}$$

At deployment, pairs follow a different distribution Π' with size marginal Ψ'.

In the following, we study PFNs through the lens of risk minimization. Risks average over a task, a context $\mathcal{D}_n \sim p^{\otimes n}$ and an independent query $(\mathbf{X}, Y) \sim p$. We write $\mathbb{E}_p[\cdot] := \mathbb{E}_{\mathcal{D}_n \sim p^{\otimes n}} \mathbb{E}_{(\mathbf{X},Y)\sim p}[\cdot]$ for a fixed pair (p, n) and $\mathbb{E}_\Pi[\cdot] := \mathbb{E}_{(p,N)\sim\Pi} \mathbb{E}_p[\cdot]$ when the pair is drawn as well, likewise for Π'. Random variables are capitalized.

2.2 Risk Under a Task Distribution

A PFN with parameters $\boldsymbol{\theta} \in \Theta$ is a measurable map $q_{\boldsymbol{\theta}} \colon (\mathbf{x}, \mathcal{D}_n) \mapsto q_{\boldsymbol{\theta}}(\cdot \mid \mathbf{x}, \mathcal{D}_n)$ into the distributions on $\mathcal{Y}$. Deployed on Π', its risk under the log loss is

$$R_{\Pi'}(q_{\boldsymbol{\theta}}) := \mathbb{E}_{\Pi'}\big[-\log q_{\boldsymbol{\theta}}(Y \mid \mathbf{X}, \mathcal{D}_N)\big]. \tag{2}$$

The following proposition characterizes the optimal predictor under an arbitrary task distribution.

Proposition 1. *Let $R_{\Pi'}(\pi_{\Pi'})$ be finite. Then for every predictor q,*

$$R_{\Pi'}(q) - R_{\Pi'}(\pi_{\Pi'}) = \mathbb{E}_{\Pi'}\big[D_{\mathrm{KL}}\big(\pi_{\Pi'}(\cdot \mid \mathbf{X}, \mathcal{D}_N) \,\big\|\, q(\cdot \mid \mathbf{X}, \mathcal{D}_N)\big)\big] \geq 0, \tag{3}$$

with equality if and only if $q = \pi_{\Pi'}$ almost surely under Π'. In particular, $\pi_{\Pi'}$ is the unique minimizer of $R_{\Pi'}$ and $R_{\Pi'}(\pi_{\Pi'})$, the Bayes risk *of Π', is the lowest risk any predictor can attain on Π'.*

The proof is given in Appendix B.1. A PFN $q_{\boldsymbol{\theta}}$ is pretrained on the population objective

$$\mathcal{L}(\boldsymbol{\theta}) := R_\Pi(q_{\boldsymbol{\theta}}). \tag{4}$$

With $\Pi' = \Pi$, Proposition 1 thus gives the pretraining objective: *a PFN is trained to minimize its risk under its own prior*, the minimizer over all predictors is π_Π (Müller et al., 2021), and the excess risk

$$\mathcal{A}_\Pi(q_{\boldsymbol{\theta}}) := R_\Pi(q_{\boldsymbol{\theta}}) - R_\Pi(\pi_\Pi) = \mathbb{E}_\Pi\big[D_{\mathrm{KL}}\big(\pi_\Pi \,\|\, q_{\boldsymbol{\theta}}\big)\big] \geq 0 \tag{5}$$

is the *amortization error*. This error is the distance, in expected KL, of the network from the PPD it is supposed to amortize.

In practice, Eq. (4) is approximated by empirical risk minimization over M datasets $\mathcal{D}^{(j)}$ whose task and size (p_j, n_j) are drawn from the prior,

$$\mathcal{L}_{\mathrm{emp}}(\boldsymbol{\theta}) := -\tfrac{1}{M} \textstyle\sum_{j=1}^M \log q_{\boldsymbol{\theta}}\big(y_j \mid \mathbf{x}_j, \mathcal{D}^{(j)}\big), \tag{6}$$

optimized with first-order stochastic gradient methods such that, in general, we have no guarantee how close we get to the target π_Π.

Figure 1 places the amortization error among the two other sources of error of a deployed PFN: the *context error*, incurred even by π_Π because a finite context leaves the task uncertain, and the *prior error*, incurred because real tasks follow $\Pi' \neq \Pi$. We focus on the former and work towards its precise relation to the other two in Proposition 2. As a first step, we propose how to analyse the amortization error for a tractable but realistic reference prior in the following section.

2.3 Measuring Amortization Error

Measuring $\mathcal{A}_\Pi(q) = \mathbb{E}_\Pi[D_{\mathrm{KL}}(\pi_\Pi \| q)]$ directly requires knowledge of the true PPD, which is intractable for the priors used by current tabular foundation models. The evidence that trained PFNs closely track the PPD they amortize comes from Müller et al. (2021, 2024). The considered settings are low-dimensional compared to current structural causal model (SCM) prior (Qu et al., 2026; Hollmann et al., 2025), none use a current PFN architecture at scale, and only small context sizes